ENGINEERS LEVEL · SEMESTER 1

Password Strength Checker

Student Workbook

Python functions and conditions — used to understand what actually keeps you safe online.

Name: ______________________________

THEORY

Python Functions + What Keeps You Safe

Python defines functions with `def` and uses **indentation** for the body. `if / elif / else` chooses among options. A strength checker measures **length** (`len`) and variety. The cybersecurity core: long, **unique** passwords resist guessing, and **hashing** lets a site verify you without storing your real password.

> Discuss: why is a long passphrase stronger than a short complex password, and why should sites never store passwords in plain text?

Trace it

Read the Python and give the result.

```
def strength(pw):
    if len(pw) >= 12:
        return "strong"
    return "weak"
```

What does strength("apple") return? ______

Code we use:
def (function) | if / elif / else | len( ) | string methods | input( ) | return

Words we are learning

- **Function** — reusable code defined with `def`; indentation marks the body.
- **Condition** — `if / elif / else` chooses a path.
- **String method** — e.g. `len(pw)` for length; checks for digits/symbols.
- **Password strength** — length + variety make guessing far harder.
- **Hashing** — a one-way scramble; sites store the hash, not your real password.

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Functions with def

Python uses def and **indentation**. Trace the code, then write a small function.

```
def greet(name):
    return "Hi " + name
```

What does greet("Sam") return? _____

Write a function that returns the length of a password:

```
def ______(pw):
    return len(pw)
```

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If / Elif / Else

Strength has tiers. Complete the conditions so longer passwords score higher.

```
if len(pw) >= 12: return "strong"
```

```
elif len(pw) >= 8: return "okay"
```

```
____: return "weak"
```

Fill the blank so short passwords are 'weak':

else

if

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What Makes A Strong Password

Cybersecurity: rate each password and note why.

12345	very weak
correct-horse-battery	strong passphrase
P@ss1	short, reused everywhere = weak

Two rules for strong, safe passwords:

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Plan My Checker + Hashing

Plan your checker and explain hashing in your own words.

My strength rules (length / variety):

In one sentence, what does hashing do?

sketch the checker UI

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PROJECT

Password Strength Checker — Plan

> Build a checker that takes a password as input and rates its strength with a Python function and if/elif/else — then explain why length and uniqueness matter, and what hashing does.

1. Inputs and output:

---

2. My function:

```
def strength(pw):
```

```
    ...
```

3. My strength tiers (if/elif/else):

---

4. My one-line explanation of hashing:
